Decoding Spoken Words from Stereotactic EEG: Linear and Transformer-Based Spectrogram Reconstruction

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Abstract

Speech neuroprostheses aim to restore communication by decoding speech from neural activity. While most existing approaches rely on electrocorticography (ECoG) or microelectrode arrays (MEA), stereotactic EEG (sEEG) remains comparatively under-explored despite its widespread use in clinical epilepsy monitoring. In this work, we investigate speech spectrogram reconstruction from sEEG recordings using the publicly available Dutch word-production dataset including 10 participants. We reproduce the dataset's linear regression baseline and compare it with a Transformer encoder, while evaluating the influence of temporal context and PCA-based dimensionality reduction. Without PCA, the Transformer consistently outperforms linear regression, reaching a mean Pearson correlation of 0.769 with an 8-frame context window compared with 0.676 for the baseline. Increasing temporal context improves Transformer performance, while PCA-based compression decreases performance, suggesting that preserving the full electrode representation is beneficial for this type of model. These results provide further evidence that Transformer-based approaches can effectively model sEEG features for speech decoding and motivate future work on more relevant evaluation metrics and cross-participant generalization.

1 Introduction

Speech production is a highly coordinated process involving several neural systems. During speech production, neural populations in regions such as the ventral sensorimotor cortex and middle precentral gyrus generate motor commands that are transmitted through the corticobulbar pathway to the muscles involved in articulation [1]. When these pathways are damaged, speech can become severely impaired. Diseases include dysarthria and anarthria [1]. In this context, speech neuroprostheses seek to restore communication in order to decode intended speech directly from brain activity. Recent studies have demonstrated that intracranial recordings can be translated into words with increasing accuracy [1]. However, current high-performing systems primarily rely on recording tools such as electrocorticography (ECoG) [2, 3] and microelectrode arrays (MEA) [4]. These modalities provide high-quality signals but offer limited spatial coverage.

Stereotactic EEG (sEEG) represents an alternative approach [5]. Used in pharmaco-resistant epilepsy patients, sEEG involves the implantation of thin depth electrodes that sample neural activity across multiple cortical and subcortical regions. Its spatial sampling is less dense than ECoG. However, its ability to record from distributed brain regions make it a promising source of data for developing more robust speech decoding models. Nevertheless, sEEG-based speech reconstruction remains relatively under-explored compared with other intracranial modalities [6, 7, 8].

Regarding the automatic modeling of these signals; recent studies have applied deep learning models to speech decoding from neural activity. Among these models, Transformers have attracted interest because they can capture long-range dependencies through self-attention [9]. In the context of sEEG-based speech reconstruction, Wu et al. [10] showed that Transformer and recurrent neural network (RNN) models outperform a linear regression baseline on the Verwoert et al. dataset [11]. However, the influence of temporal context and neural feature representation on Transformer performance has not been investigated in detail.

In this work, we investigate speech spectrogram reconstruction from sEEG recordings using the publicly available Dutch word-production dataset introduced by Verwoert et al. [11]. We reproduce the dataset's linear regression baseline and compare it with a Transformer encoder trained on the same task. We further evaluate the impact of temporal context and PCA-based dimensionality reduction on decoding performance, and analyze electrode importance using SHAP to identify the neural features contributing to model predictions.

2 Related Work

Early work on neural speech decoding primarily focused on ECoG recordings. Anumanchipalli et al. [2] proposed a two-stage bidirectional LSTM that first predicts articulatory movements from neural activity before synthesizing speech. In addition, Makin et al. [3] formulated speech decoding as a sequence-to-sequence translation problem using recurrent neural networks. Regarding MEA, Moses et al. [4] later demonstrated real-time speech decoding in a participant with anarthria.

Compared with ECoG and MEA, relatively few studies have investigated speech reconstruction from sEEG recordings. Angrick et al. [6] proposed a convolutional neural network that processes each electrode shaft independently before combining the extracted features for speech reconstruction. Kohler et al. [7] employed an encoder-decoder architecture for speech reconstruction from depth-electrode recordings, while Wang et al. [8] investigated the cortical dynamics underlying speech production using intracranial recordings.

More recently, Wu et al. [10] benchmarked linear regression, recurrent neural networks, and Transformer models on the Verwoert et al. dataset [11], reporting that the deep learning models consistently outperformed the linear baseline. The present work extends this benchmark by studying the influence of temporal context, PCA-based dimensionality reduction, and model interpretability through SHAP-based electrode importance analysis.

3 Data and Methods

3.1 Dataset

We use the iBIDS dataset of Verwoert et al. [11]. sEEG signals were collected from 10 native Dutch participants with pharmaco-resistant epilepsy. Participants read aloud 100 Dutch words from the Dutch IFA corpus [12] and the numbers from one to ten. Each word was shown for 2 seconds, followed by a fixation cross during 1 second. Each recording lasted around 300 seconds per participant. Neural data was obtained with clinically implanted platinum-iridium sEEG electrode shafts. The number of electrodes per participant ranged from 54 to 127. A more detailed description of the data is available in the original paper [11].

3.2 Feature Extraction

Feature extraction was performed using the official preprocessing pipeline provided with the dataset. High-gamma features ranging from 70 to 170 Hz were extracted from the raw sEEG signals using a 4th-order IIR bandpass filter. It was followed by two IIR bandstop filters to remove the 50 Hz line noise and its first harmonic. The Hilbert envelope of the signals was computed and averaged over 50 ms windows with a 10 ms frame shift, yielding one feature vector per electrode every 10 ms. To incorporate temporal information, each feature vector was concatenated with neighboring frames, corresponding to temporal context ranging from ±100 ms to ±400 ms corresponding to model orders 2, 4, 6, 8 (i.e. the number of preceding and following 50 ms frames). Therefore, the shape of input vectors is $(2 \cdot \text{order} + 1) \times n_{\text{electrodes}}$. Neural features were normalized to zero mean and unit variance using the mean and standard deviation computed on the training data for each cross-validation fold. The same transformation was then applied to the corresponding test data. Audio recordings were downsampled to 16 kHz and converted into Short-Time Fourier Transform (STFT) spectrograms using the same window and frame shift. Finally, it was transformed into 23-bin log-mel spectrograms, the prediction targets.

3.3 Model Architecture and Training

To explore the effectiveness of a Transformer architecture for speech decoding using sEEG, we compare it to a linear regression baseline. To further study the influence of dimensionality reduction, we trained every model with and without PCA on the first 50 principal components.

Linear Regression. Linear Regression serves as the baseline model provided with the dataset. The model predicts the 23-bin log-mel spectrogram from the stacked high-gamma neural features. Each model was estimated independently using a least-squares approach for each participant. Following the original work, the baseline was evaluated using 10-fold cross-validation.

Transformer. The Transformer was chosen because self-attention allows interactions between neural activity recorded at different temporal positions to be learned directly. We implemented a Transformer encoder for speech spectrogram reconstruction using the same neural features as the linear regression baseline. The input feature vector is first reshaped into a sequence of temporal frames, where each frame corresponds to the activity recorded across all electrodes at a given time step. Each frame is projected from the electrode space to a 128-dimensional embedding through a linear layer, after which fixed sinusoidal positional encodings [9] are added to preserve the temporal order of the sequence.

The embedded sequence is processed by a Transformer encoder composed of three encoder layers. Each layer consists of a multi-head self-attention module with four attention heads followed by a position-wise feed-forward network of dimension 256. The resulting sequence representations are aggregated using global mean pooling across the temporal dimension and passed through a two-layer multilayer perceptron with a ReLU activation to predict the 23-bin log-mel spectrogram.

Models were trained independently for each participant using 5-fold cross-validation, due to higher computational cost of training than with linear regression. Within each training fold, 10% of the training samples were randomly selected as a validation set for early stopping, with a patience of 10 epochs. The model was optimized using the Adam optimizer with a learning rate of 10^{-3} and a batch size of 128 for a maximum of 50 epochs. Mean squared error (MSE) between the predicted and reference log-mel spectrograms was used as the training loss. The model achieving the lowest validation loss was retained.

3.4 Audio Reconstruction

Since log-mel spectrograms do not contain phase information, they cannot be converted directly into audio. To qualitatively evaluate the reconstructed speech, we used the Griffin-Lim algorithm [13]. This method iteratively estimates the missing phase to reconstruct an audio waveform from the predicted log-mel spectrogram.

3.5 Evaluation

Following the dataset paper, performance was evaluated using the Pearson correlation coefficient between the predicted and

Table 1: Grand mean Pearson correlation across 10 participants, by temporal context (number of stacked frames) and model, without PCA.

Model	2	4	6	8
Linear regression	0.650	0.685	**0.688**	0.676
Transformer	0.652	0.724	0.752	**0.769**

the reference log-mel spectrogram, averaged across mel bins and cross-validation folds, then across participants for grand mean.

3.6 Model Interpretability

To identify the electrodes that contributed most to the predictions, we performed a feature importance analysis for both models. For the linear regression baseline, electrode importance was derived directly from the learned model coefficients, trained without PCA. The absolute values of the regression coefficients were aggregated across output mel bins and temporal offsets to compute a single importance score per electrode. For the Transformer, electrode importance was estimated using the SHAP GradientExplainer. The explainer was configured with 50 background samples, 30 explained samples, and 50 Monte Carlo samples per explanation. Absolute SHAP values were averaged across output mel bins, explained samples, and temporal offsets to obtain a single importance score for each electrode. The three electrodes with the highest importance scores were then reported for each participant.

4 Results

This section presents the decoding performance of the Transformer model compared with the linear regression baseline, the effect of PCA-based dimensionality reduction, and the analysis of electrode importance.

4.1 Linear Regression vs. Transformer

Table 1 reports the mean Pearson correlation coefficient obtained by the linear regression baseline and the Transformer across the four temporal context sizes without PCA. In general terms, both approaches improved their decoding performance as the temporal context increased, although with a stronger benefit for the Transformer-based architecture. More specifically, the linear regression baseline achieved its highest mean correlation at a context of 6 frames (0.688), whereas the Transformer continued to improve up to a context of 8 frames (0.769).

Performance also varied considerably across participants (Figure 1). At a context of 8 frames, the highest Transformer correlations were obtained for participants *sub-06* ($r = 0.89$) and *sub-03* ($r = 0.86$), whereas *sub-01* and *sub-05* remained below $r = 0.62$. No clear relationship was observed between decoding performance and the number of implanted electrodes.

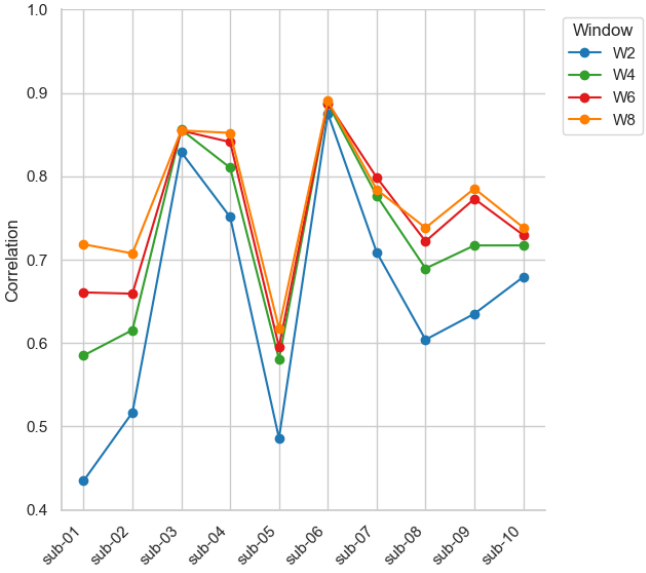


Figure 1: Mean Pearson Coefficient Correlation across folds for each participant at all context levels.

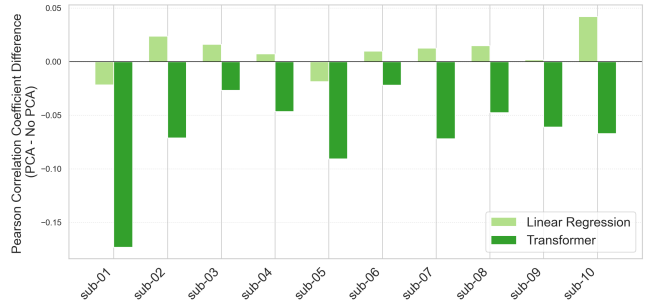


Figure 2: Effect of PCA on Spectrogram Reconstruction with temporal context of 4 frames.

4.2 Effect of Principal Component Analysis

To evaluate the influence of dimensionality reduction, both models were trained with and without PCA. Figure 2 summarizes the results for our models at a context size of 4 frames, which is the number of frames used in the original paper [11].

Applying PCA consistently reduced Transformer decoding performance across all participants (Figure 2). Without PCA, the Transformer achieved a grand mean Pearson correlation of 0.724 (at a 4-frame context), whereas performance dropped by 0.068 down to 0.656 when PCA was applied. Conversely, PCA slightly improved the grand mean performance of the linear regression. We observe an increase of 0.009, from 0.685 to 0.694. The overall performance comparison across configurations is summarized in Table 2.

Table 2: Grand mean Pearson correlation across all 10 participants with and without PCA (temporal context = 4 frames).

Model	No PCA	PCA (50 components)
Linear Regression	0.685	**0.694**
Transformer	**0.724**	0.656

4.3 Electrode Importance and Anatomical Localization

Comparing the top three electrodes identified by the two models showed little overlap for most participants, indicating that the Transformer and the linear regression baseline relied on different sets of neural features. In both models, feature importance values were relatively uniform across the highest-ranked electrodes, with no single electrode consistently dominating predictions. To identify anatomical patterns shared across participants, the three most important electrodes for each participant were mapped to their corresponding Destrieux atlas labels.

Linear Regression. For 6 out of 10 participants (sub-01, sub-04, sub-05, sub-07, sub-08, and sub-10), the top-ranked electrode identified by the linear regression model remained identical across all four context window sizes (2, 4, 6, and 8 frames). For the remaining participants, the highest-ranked electrode shifted only between adjacent contacts on the same depth shaft, indicating that electrode importance was largely stable across temporal contexts.

Evaluating the linear regression baseline at its optimal context size of 6 frames, many top-ranked electrodes mapped to non-cortical or unlabeled regions (Unknown, $n = 11$; Left cerebral white matter, $n = 6$; Right cerebral white matter, $n = 1$). After excluding non-cortical labels, the most frequently represented cortical areas were the right middle temporal gyrus ($n = 2$), right inferior circular insular sulcus ($n = 2$), and right superior temporal sulcus ($n = 2$).

Transformer. For the Transformer model evaluated at its optimal context window of 8, the top three SHAP-ranked electrodes similarly mapped to non-cortical labels (Unknown, $n = 8$; Left cerebral white matter, $n = 6$; Right cerebral white matter, $n = 2$). After excluding these labels, the most frequently represented cortical regions were the right inferior circular insular sulcus ($n = 3$) and the right superior temporal sulcus ($n = 2$).

5 Discussion

In this study, we compared the performance of linear regression to a transformer architecture for spectrogram reconstruction using Griffin-Lim. The transformer consistently outperformed the linear regression across context windows. However, the reconstructed audio remains unintelligible. Here, we discuss our results and limitations.

5.1 On Transformer and Dimensionality Reduction

The Transformer models improve performances without PCA. However, they perform worse than the linear regression models when PCA is applied. This supports the idea that the Transformer benefits from access to the full, high-dimensional electrode representation, rather than a compressed one. This is distinct from the earlier finding on temporal context length, where the Transformer also outperformed linear regression as more context was added. While linear regression is limited to learning linear relationships between neural activity and speech features, the Transformer can exploit more complex spatial and temporal interactions across electrodes. Reducing the input to 50 principal components appears to remove information that contributes to this advantage.

5.2 Evaluation Method

Although the Transformer achieved Pearson correlations of up to 0.89 for some participants, the reconstructed speech remained largely unintelligible when converted back to audio. This observation is consistent with previous work on sEEG-based speech reconstruction [10]. It also highlights the limitation of using Pearson correlation as the sole evaluation metric. Since silent frames are generally easier to recreate, correlation values may become high even when spoken parts of speech are not accurately reproduced. Consequently, future work should complement Pearson correlation with perceptual speech-quality metrics such as Spectro-Temporal Glimpsing Index (STGI) [14], as reported in the original paper [11].

5.3 Heterogeneity of Electrode Placement

The SHAP analysis showed that brain regions contributing most to decoding differed across participants. This is likely explained by the heterogeneous placement of sEEG electrodes, which is determined by clinical needs rather than experimental design. As a result, models trained for one participant cannot be directly transferred to another.

The most important electrodes were located in white matter and unknown regions. When excluding those electrodes, it was found that some anatomical regions appeared repeatedly, including the middle temporal gyrus, superior temporal sulcus, and supramarginal gyrus although. This suggests that future work could focus on anatomical regions rather than individual electrodes. Moreover, the electrode rankings obtained with the Transformer differed from those of the linear regression baseline.

5.4 Limitations

Several limitations overshadow our work. First, the context-window and PCA experiments were not exhaustive: it would be interesting to feed the Transformer more context and do PCA experiments across all windows. Furthermore, for computational costs, 5-fold cross-validation was performed with the Transformer against 10-fold cross-validation for the Linear Regression. Besides, all results rely on Pearson correlation which is not a direct measure of speech intelligibility. Additionally, the dataset is relatively small and electrode placement is inconsistent which limits the ability to assess generalization across subjects. The dataset is also overt speech: the participant was actually producting voice and not imagining it in, which is the intent of speech neuroprostheses [15].

6 Conclusion and Future Work

In this work, we compared a Transformer encoder with a linear regression baseline for speech spectrogram reconstruction from sEEG recordings. The Transformer consistently outper-

formed the linear baseline across all temporal context sizes, while PCA-based dimensionality reduction reduced performance. Moreover, electrode importance analysis highlighted variability across participants, reflecting the heterogeneous placement of sEEG electrodes. Future work will include evaluating speech intelligibility using perceptual metrics such as STGI [14] and investigating alternative decoding tasks, such as word or phoneme classification [16]. It can also involve cross-participant model training based on shared anatomical brain regions [17].

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